

ENGINEERING STANDARD

Part Number Standard

Universal Limbs Foundation — Open-Source Prosthetic System

Document: **ULF-DOC-001**

Revision: **Rev A**

Effective: July 29, 2026

Owner: Engineering / Quality

1. Purpose & Scope

This standard defines a single, simple part-numbering system for every engineering item produced or used across the Universal Limbs Foundation prosthetic system — CAD models, STL/3MF/STEP export files, printed parts, purchased components, assemblies, drawings, and documentation. It applies to all current and future parts and is designed so that volunteers and students can assign a correct part number without engineering training.

The system intentionally avoids embedding revision, material, or configuration detail into the number itself — those are tracked separately (see Section 5) — so a part's identity never changes over its lifecycle.

2. Part Number Format



ULF	Fixed prefix — Universal Limbs Foundation. Identical on every part number.
XXX	Three-letter category code identifying the type of item (Section 3).
001	Three-digit sequential number, unique within its category, assigned in order as new items are created — never reused, never reassigned.

Example: **ULF-SKT-001** is the first socket part registered in the system. The next new socket becomes **ULF-SKT-002**, regardless of design changes to 001.

3. Category Code Table

CODE	CATEGORY	EXAMPLE
SKT	Socket	ULF-SKT-001
PLM	Palm	ULF-PLM-001
WRS	Wrist	ULF-WRS-001
THM	Thumb	ULF-THM-001
IND	Index finger	ULF-IND-001
MID	Middle finger	ULF-MID-001
RNG	Ring finger	ULF-RNG-001
LTL	Little finger	ULF-LTL-001
HRN	Harness	ULF-HRN-001
CBL	Cable	ULF-CBL-001
LNR	Liner	ULF-LNR-001
FST	Fastener (purchased)	ULF-FST-001
JIG	Assembly jig	ULF-JIG-001
TOL	Tool	ULF-TOL-001
ASM	Assembly	ULF-ASM-001
DRW	Engineering drawing	ULF-DRW-001
DOC	Document / standard	ULF-DOC-001

4. Naming Convention

- Every physical part, CAD model, assembly, drawing, and controlled document receives exactly one part number.
- Part numbers are assigned once, in a shared log, before design work begins — not derived from a filename or invented ad hoc.
- The category code describes what the item *is*, not what project or product it belongs to.
- Part numbers are permanent: they are never reused, changed, or deleted once issued, even if the part is discontinued.
- Design changes never create a new part number — they create a new revision of the same part number (Section 5).

5. File Naming Convention

Every exported file is named with its part number and revision, separated by an underscore, keeping the original file extension:

FILE TYPE	FILE NAME
CAD (Fusion 360)	ULF-SKT-001_RevB.f3d
STL	ULF-SKT-001_RevB.stl
3MF	ULF-SKT-001_RevB.3mf
STEP	ULF-SKT-001_RevB.step

Never include dates, initials, or words like "final" / "latest" in a file name — the part number and revision are the only identifiers a file needs.

6. Revision Rules

- Revision is **never** part of the part number — it is tracked as a separate field everywhere the part number appears.

- Use sequential letters: **Rev A** , **Rev B** , **Rev C** ... (numeric shops may instead use **Rev 01** , **Rev 02** , **Rev 03** — pick one convention and use it project-wide).
- Every new revision must have a documented reason (fit issue, strength change, manufacturability, etc.) recorded in the part's change log.
- Rev A is the first released revision; unreleased work-in-progress design is not yet a controlled revision.

Build Records display all three fields together: **Part Number** **ULF-SKT-001** · **Revision** **Rev B** · **Print Attempt** **2** .

Part number identifies *what* the part is. Revision identifies *which version* of its design. Print attempt identifies *which physical build* of that revision this record documents.

7. Assemblies & Purchased Components

PART NUMBER	ITEM	TYPE
ULF-ASM-001	Universal Limbs Hand	Assembly
ULF-FST-001	Socket head screw, M3×10	Purchased

- Assemblies get their own part number (**ASM**) — separate from any single part inside them — and list their child part numbers on the assembly drawing or BOM.
- Purchased, off-the-shelf items (screws, bearings, cable, adhesive) also receive an internal part number so they can appear on drawings and BOMs like any other item, even though they are not designed in-house.

8. Example Part Number List

PART NUMBER	DESCRIPTION
ULF-SKT-001	Transradial socket — Rasna configuration
ULF-PLM-001	Palm structure
ULF-THM-001	Thumb linkage
ULF-HRN-001	Harness strap assembly
ULF-CBL-001	Actuation cable
ULF-ASM-001	Universal Limbs Hand (full assembly)
ULF-DRW-001	Socket engineering drawing

9. Rules for Future Expansion

- New category needed? Propose a clear, memorable three-letter code and add it to the master Category Code Table (Section 3) — do not create a category outside this table.
- Never reuse a retired category code or a retired sequence number for a different item.
- If a category exceeds 999 items, extend the sequence to four digits (e.g. **ULF-FST-1000**) rather than inventing a new code.
- All part number assignments are logged in one shared register (spreadsheet or PLM tool) so no two people can accidentally issue the same number.
- Changes to this standard itself follow the same revision process — this document is **ULF-DOC-001** and any change to these rules ships as a new revision, not a new document.